PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Protected entry still missing; OV chopper and fault memory populated on dedicated pages.

GEN DRV8353S driver and three low-side current channels



File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only



File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels



File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only



File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries



File: external-boundaries.kicad_sch

Bus-fed auxiliary and independent safety supply



File: aux-supply.kicad_sch

Autonomous OV chopper comparator and rail / brake monitors



File: protection-comparators.kicad_sch

Release-before-rearm DFF memory



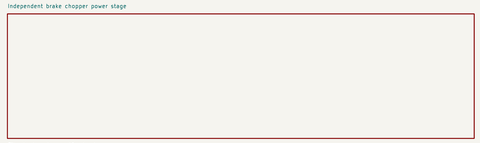
File: fault-latch.kicad_sch

LVC Schmitt fault inputs without series diode offsets



File: fault-interfaces.kicad_sch

Independent brake chopper power stage



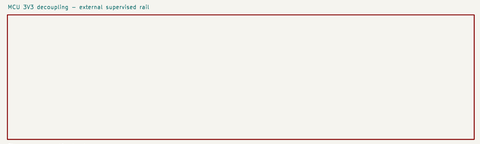
File: brake-chopper.kicad_sch

STM32G474VET6 LQFP100: power, reset, debug and control



File: mcu-core.kicad_sch

MCU 3V3 decoupling — external supervised rail



File: mcu-power.kicad_sch

SW CSA ADC input filters and brake status sense



File: mcu-analog.kicad_sch

3V3 Hall interfaces and optional quadrature encoder



File: mcu-sensors.kicad_sch

GEN independent raw CSA window trips



File: gen-fast-oc.kicad_sch

HUB independent raw CSA window trips



File: hub-fast-oc.kicad_sch

Ratiometric OC window and wired sink fault



File: fast-oc-reference.kicad_sch

Bounded bench DC entry — manual dead-circuit sequencing only



File: dc-entry.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheets: /

File: powerboard-rev4.kicad_sch

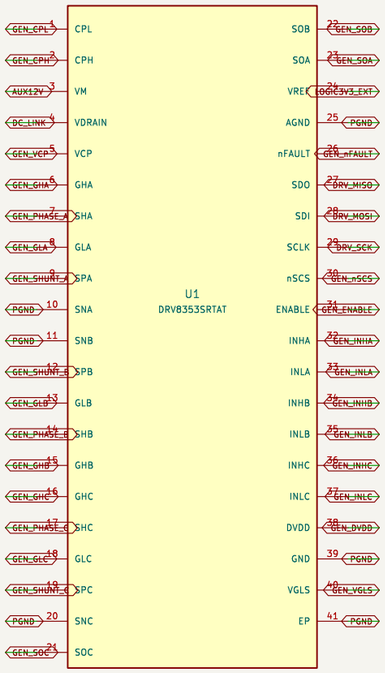
Title: Pedal-by-wire: hierarchical engineering review

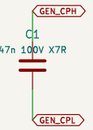
Size: A1	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 1/19

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

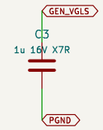
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.



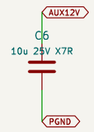






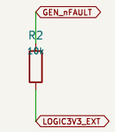


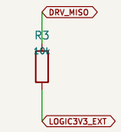


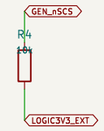












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

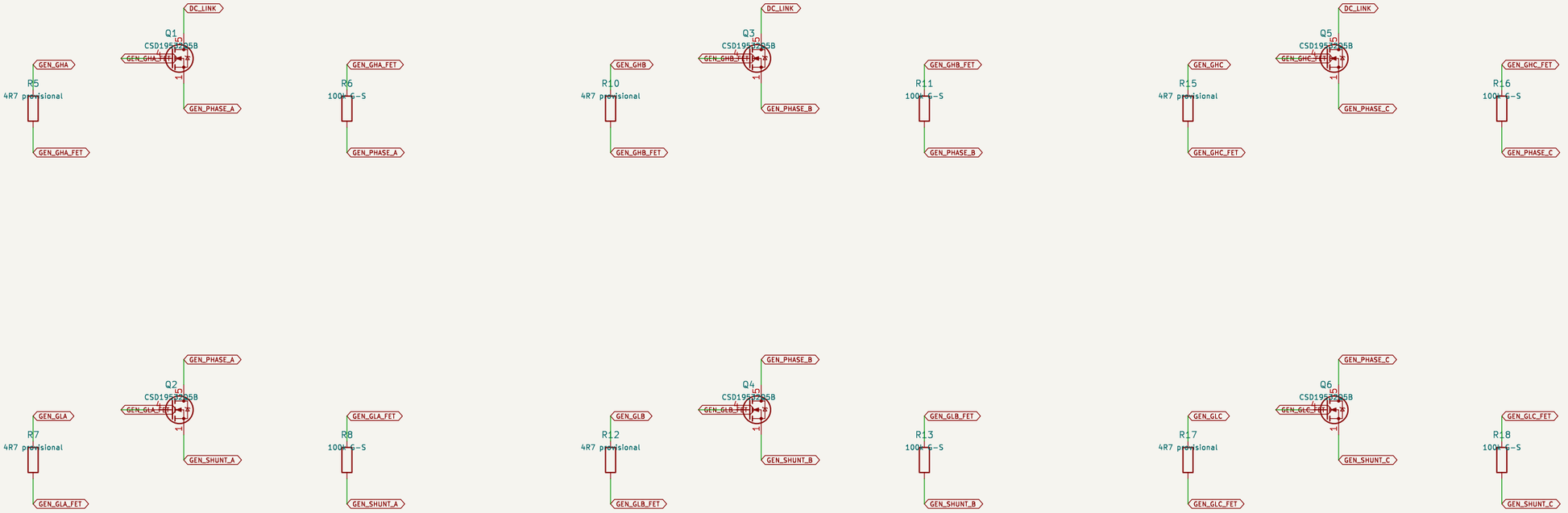
Sheet: /GEN DRV8353S driver and three low-side current channels/
File: gen-driver.kicad_sch

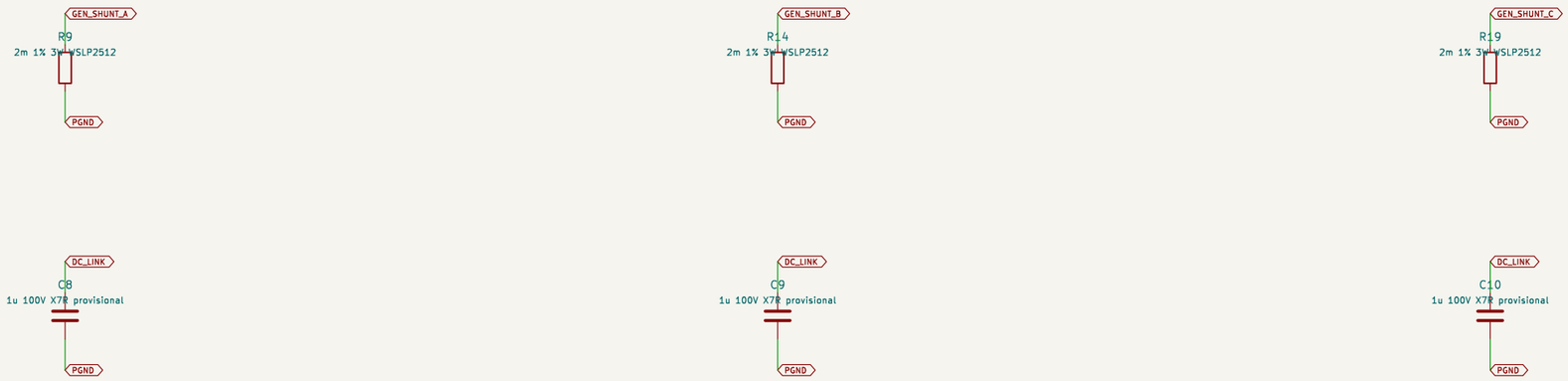
Title: GEN DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 2/19

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.





Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

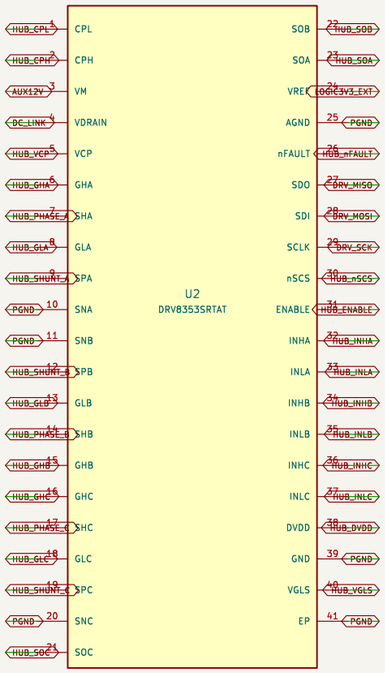
Title: GEN 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 3/19

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

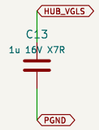
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.

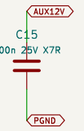






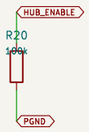


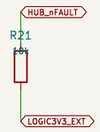


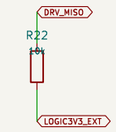


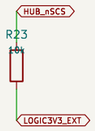












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

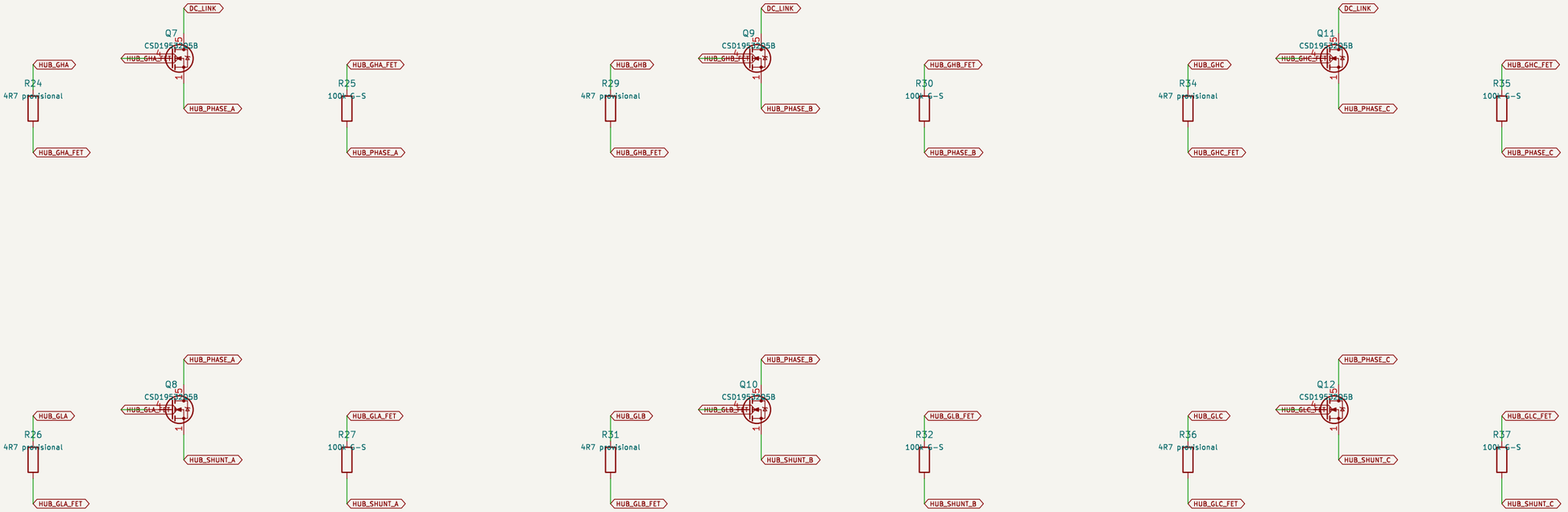
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

Title: HUB DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 4/19

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

Title: HUB 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 5/19

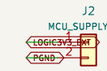
OFFBOARD interfaces: 3V3 supply, brake loop and protected bus entry.

DO NOT ENERGIZE. Protected entry unfinished; MCU and protection remain unqualified.

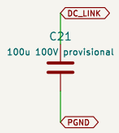












Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Populated: OV chopper, fault latch, brake loop, watchdog, aux supply. Added: raw CSA window trip; timing/coverage unqualified.

MCU and sensor interfaces populated; firmware, thermal sensors, storage/BMS acceptance missing.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

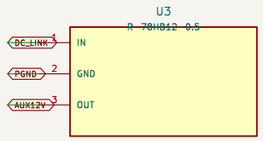
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

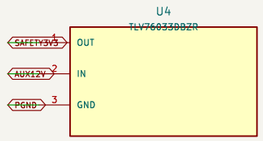
Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

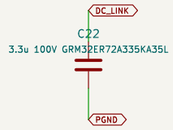
Title: Unfinished safety / control / supply boundaries

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 6/19

R-78HB12-0.5: 17–72V operating, 75V absolute. Entry transient protection still REQUIRED.



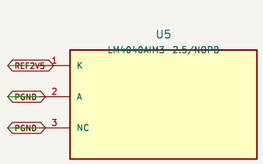




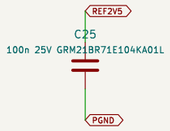












Safety supply is NOT the MCU supply. LOGIC3V3_EXT remains a supervised, external MCU rail.

Supply total/current/thermal budget provisional; no galvanic isolation or stored-energy ride-through.

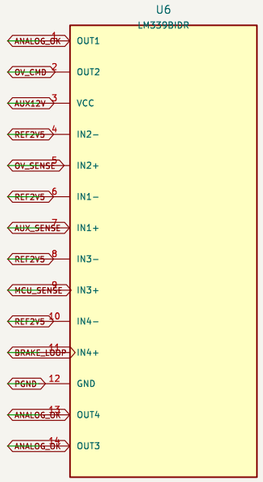
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

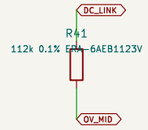
Sheet: /Bus-fed auxiliary and independent safety supply/
File: aux-supply.kicad_sch

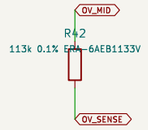
Title: Bus-fed auxiliary and independent safety supply

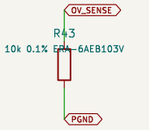
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 7/19

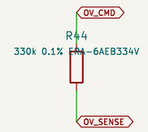
Comparator powered from AUX12V; open collectors pulled to SAFETY3V3. No MCU in OV path.



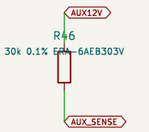


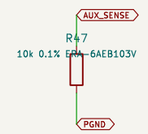


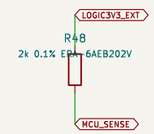


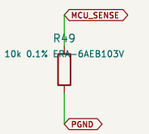


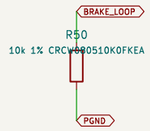




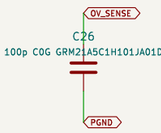














BRAKE_LOOP must close to SAFETY3V3 through normally-closed brake + thermal switches. Open wire trips.

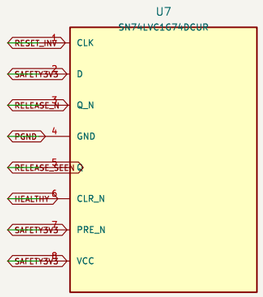
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

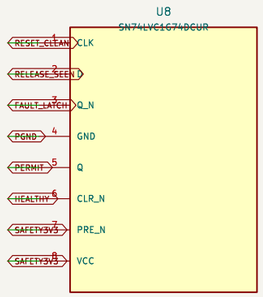
Sheet: /Autonomous OV chopper comparator and rail / brake monitors/
File: protection-comparators.kicad_sch

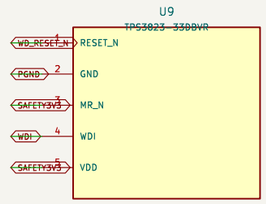
Title: Autonomous OV chopper comparator and rail / brake monitors

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 8/19

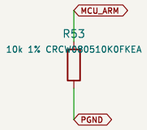
UNSAFE DRAFT: healthy release edge THEN press edge required. No automatic reset.













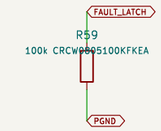


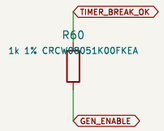


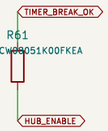


















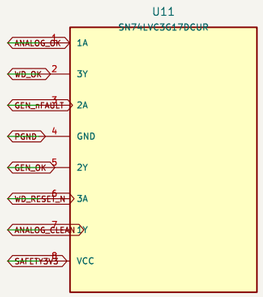
TPS3823 non-A: falling WDI only; hold low during RESET. Stuck RESET stays disabled.

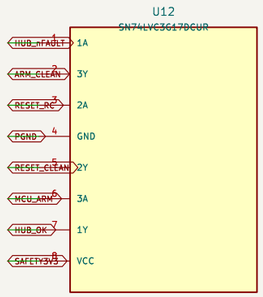
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

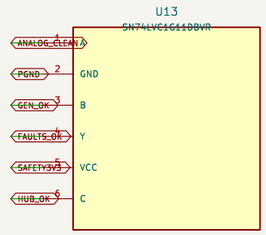
Sheet: /Release-before-rearm DFF memory/
File: fault-latch.kicad_sch

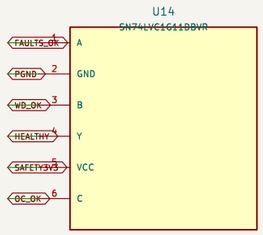
Title: Release-before-rearm DFF memory

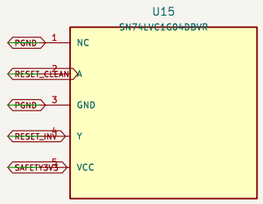
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 9/19

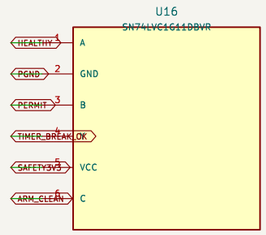
























Default OFF via async supervisor clear. Analog ramp/metastability NOT validated.

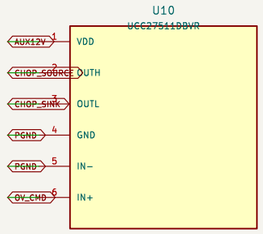
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

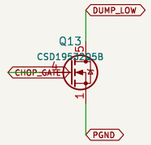
Sheet: /LVC Schmitt fault inputs without series diode offsets/
File: fault-interfaces.kicad_sch

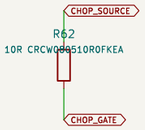
Title: LVC Schmitt fault inputs without series diode offsets

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 10/19

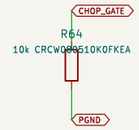
Autonomous OV path ignores bridge ENABLE. External resistor is required across BRAKE_DUMP_RESISTOR connector pins.

















Candidate: 2 x TE HSC3004R7J in series (9.4 ohm). Chassis/heatsink + NC thermal cutoff required.

No resistor cable clamp / fuse / SOA qualification yet. Loss of AUX supply loses bus sink: NOT FAIL SAFE.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

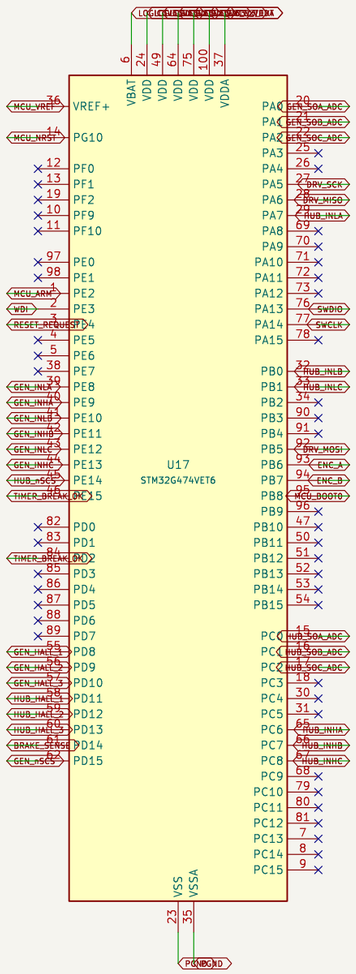
Sheet: /Independent brake chopper power stage/
File: brake-chopper.kicad_sch

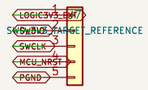
Title: Independent brake chopper power stage

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 11/19

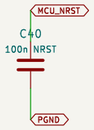
HSI16 internal clock / PLL option: no HSE or LSE fitted; PF0/PF1 and PC14/15 explicit NC.

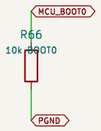
BOOT0 low; keep NRST function and configure boot option bytes before use. No firmware supplied.











BKIN active LOW required in TIM1/TIM8. Separate hardware ENABLE path remains intact.

NC GPIO: leave analog mode; HSI accuracy / PLL / ADC timing require validation.

MCU_ARM / RESET_REQUEST 10k and WDI 1k default-low resistors are on fault-latch page.

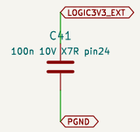
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

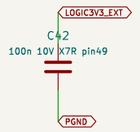
Sheet: /STM32G474VET6 LQFP100: power, reset, debug and control/
File: mcu-core.kicad_sch

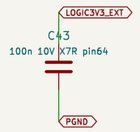
Title: STM32G474VET6 LQFP100: power, reset, debug and control

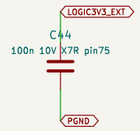
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 12/19

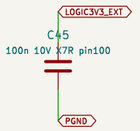
One 100n per VDD pin (24,49,64,75,100) plus VBAT pin6. Place each at pin / ground return.

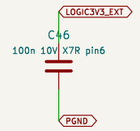


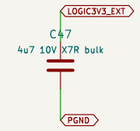


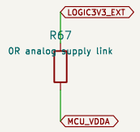


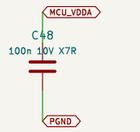


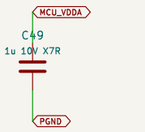


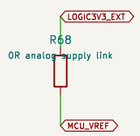


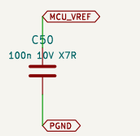


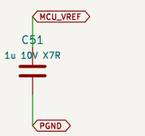












VREFBUF MUST remain disabled: external 3V3 reference. VDDA/VREF tied via OR; not precision isolated.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

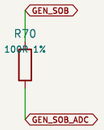
Sheet: /MCU 3V3 decoupling — external supervised rail/
File: mcu-power.kicad_sch

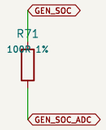
Title: MCU 3V3 decoupling — external supervised rail

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 13/19

100R / 1nF candidate filters: settling / sample aperture / CSA gain remain unqualified.

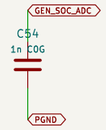




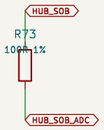


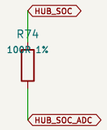




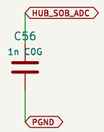


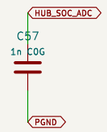


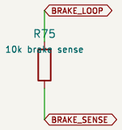


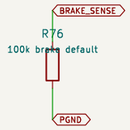


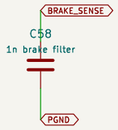












BRAKE_SENSE is status ONLY. Brake comparator and fault path do not depend on firmware.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Six CSA ADC input filters and brake status sense/
File: mcu-analog.kicad_sch

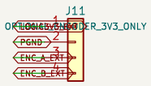
Title: Six CSA ADC input filters and brake status sense

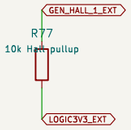
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 14/19

3V3 ONLY: open-collector Hall / encoder or compatible 3V3 outputs. NO 5V tolerance claim.



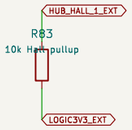






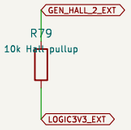






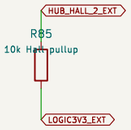






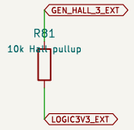




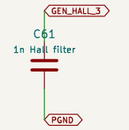


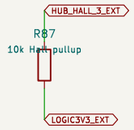






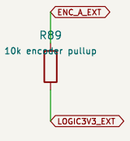




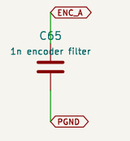


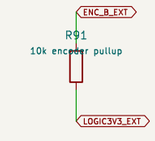


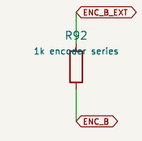


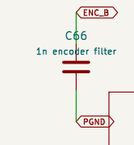












Encoder PB6/PB7: TIM4_CH1/CH2 AF2 DS12288r5 T13 p74. Cable ESD/EMC protection unresolved.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

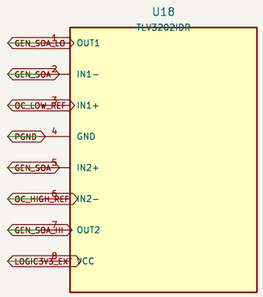
Sheet: /3V3 Hall interfaces and optional quadrature encoder/
File: mcu-sensors.kicad_sch

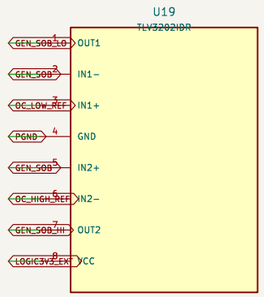
Title: 3V3 Hall interfaces and optional quadrature encoder

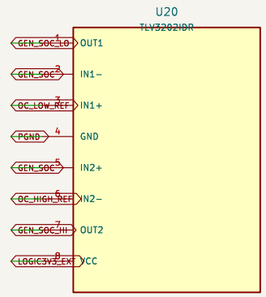
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 15/19

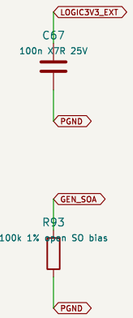
TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

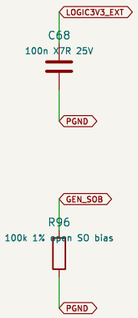
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.



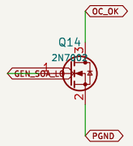


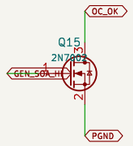


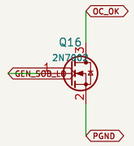


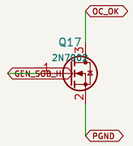


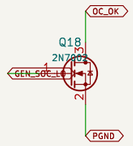


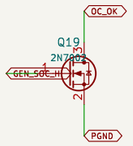


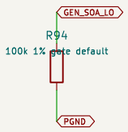


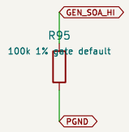


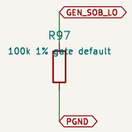


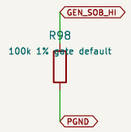


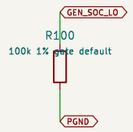


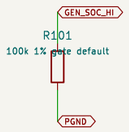












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

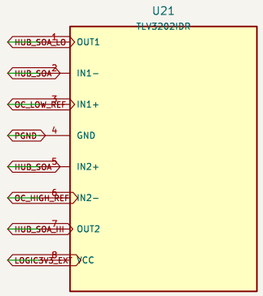
Sheet: /GEN independent raw CSA window trips/
File: gen−fast−oc.kicad_sch

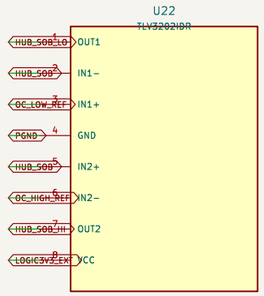
Title: GEN independent raw CSA window trips

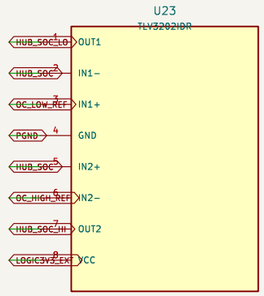
Size: A3	Date:	**Rev: A−PARTIAL**
KiCad E.D.A. 9.0.9		Id: 16/19

TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

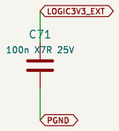
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

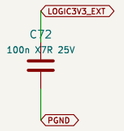




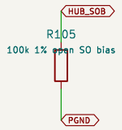


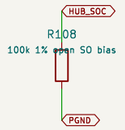


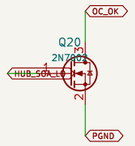


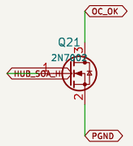


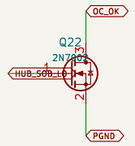


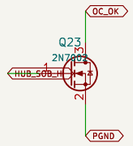


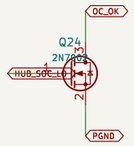


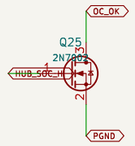


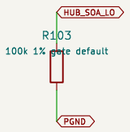


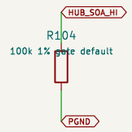


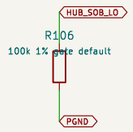


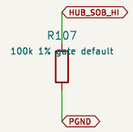


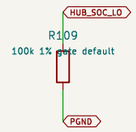


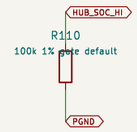












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE − SAFETY CIRCUITS UNFINISHED

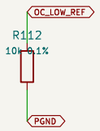
Sheet: /HUB independent raw CSA window trips/
File: hub−fast−oc.kicad_sch

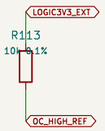
Title: HUB independent raw CSA window trips

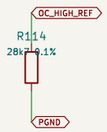
Size: A3	Date:	Rev: A−PARTIAL
KiCad E.D.A. 9.0.9		Id: 17/19

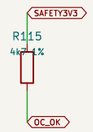
VREF tracks actual driver LOGIC3V3_EXT: 0.852713 / 2.447287 V nominal at 3.3 V.











OC_OK directly enters U14.6, clears U7/U8 and U16: both ENABLE and TIM1/TIM8 break.

No external hysteresis or blanking: first crossing latches. Noise/settling and sink delay unqualified.

Nominal trip +/−19.932 A; provisional measurement +/−30 A. Commissioning cap 10 A, NOT release.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

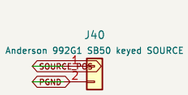
Sheet: /Ratiometric OC window and wired sink fault/
File: fast-oc-reference.kicad_sch

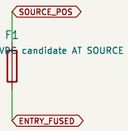
Title: Ratiometric OC window and wired sink fault

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NOT SAFE TO ENERGIZE integrated drive. No automatic sequencing or reverse-blocking guarantee.

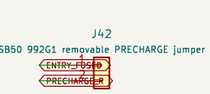
36-58.4V bench interface, 15A TARGET ONLY. Disconnect PM machines before source removal.

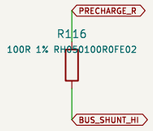


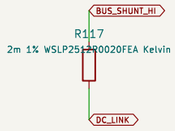


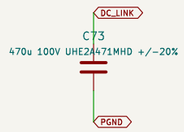


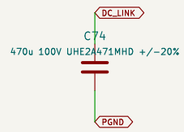




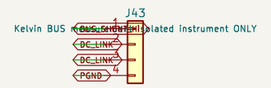












J41 closes only after measured |SOURCE-BUS| <1V; J42 then removed. NO hot unplug.

J43 raw high common mode: NOT MCU ADC. D20/fuse reverse-fault coordination UNQUALIFIED.

Sink/chopper stays DC_LINK side. MAIN open does NOT make bus safe: PM backfeeds via diodes.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /Bounded bench DC entry – manual dead-circuit sequencing only/
File: dc-entry.kicad_sch

Title: Bounded bench DC entry – manual dead-circuit sequencing only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 19/19